

Project Name:	LaunchPad AI	Team Members:	Chanchal Patni, Shilpa Kumari
Max Marks:	2 Marks		

Data Flow Diagram (DFD) Specification

This document models how information moves logically through the **LaunchPad AI** environment—tracing data transitions between external users, backend handling algorithms, data stores, and external Generative LLM application programming interfaces.

1. SYSTEM ENTITIES & COMPONENTS MAPPING

Component Type	System Element Name	Description / Role within LaunchPad AI
External Entity	User / Entrepreneur	Provides startup business descriptions and configuration options via the web frontend. Receives structured validation outputs.
External Entity	Google Gemini AI API	External foundational engine that processes structured text prompts to compute validation metrics and SWOT text.
Process 1	Request Handling & Input Validation	Inbound Flask controller checking text limits, checking inputs for faults, and generating operational acknowledgements.
Process 2	AI Prompt & Analysis Orchestration	Constructs system business templates and context envelopes to dispatch cleanly to the Gemini 1.5 Flash backend.
Process 3	Dynamic Dashboard Rendering	Transforms unstructured raw JSON/markdown responses from the model into functional UI cards and visualization segments.
Data Store	System Configuration / SQLite DB	Maintains app dependencies, basic transactional logs, or user session attributes.

2. DATA FLOW SEQUENCE & TRANSITIONS

Flow ID	From (Source)	To (Destination)	Data Label / Information Passed
DF-1	User / Entrepreneur	Process 1: Request Handling	Raw text startup concepts, market sector constraints, and submission request payloads.
DF-2	Process 1: Request Handling	User / Entrepreneur	Instant request status acknowledgement or form validation error alerts.
DF-3	Process 1: Request Handling	Data Store: System Configuration	Sanitized session criteria and tracking properties.
DF-4	Process 1: Request Handling	Process 2: AI Prompt Orchestration	Clean parsed text attributes containing concept specifications.
DF-5	Process 2: AI Prompt Orchestration	External Entity: Google Gemini API	Fully constructed business analytical prompt structures (.env secure token added).
DF-6	External Entity: Google Gemini API	Process 2: AI Prompt Orchestration	Raw viability indicators, structured SWOT payload responses, and commercialization matrices.
DF-7	Process 2: AI Prompt Orchestration	Process 3: Dynamic Dashboard Rendering	Unfiltered business intelligence matrices and metric scores.
DF-8	Process 3: Dynamic Dashboard Rendering	User / Entrepreneur	Polished interactive interface summaries, dynamic SWOT reports, and Ngrok-mapped view updates.